

ARTICLE

The efficient phasing and imputation pipeline of low-coverage whole genome sequencing data using a high-quality and publicly available reference panel in cattle

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Abstract

Low-coverage whole genome sequencing (lcWGS) has great potential to effectively genotype large-scale population and to provide solid data for imputation; however, the time for imputation needs to be optimized. There is also no publicly available reference panel for whole genome selection in cattle. Here, we proposed a combination of Beagle 5.4 for phasing and GLIMPSE2 for imputation, which is fast and accurate for cattle lcWGS data. Furthermore, we established a multi-breed reference panel with 61.8 million SNPs based on 2976 worldwide cattle, of which 1766 were bulls, by evaluating diversity and the size of the reference panel. The evaluation of imputation accuracy was conducted using new reference panel for both lcWGS and Bovine BeadChip data. The average concordance rate in Holstein was 99.6%, 99.6%, and 99.5% for 1X, 0.5X, and 0.1X lcWGS data, 99.5% and 99.0% for 777K and 50K chip data, and it was 98.8% for 1X lcWGS data in Simmental. We further investigated the factors affecting the imputation accuracy of lcWGS data and discovered that segmental duplication, structural variant, and guanine-cytosine content were the top three factors. Interestingly, we found that 10 regions longer than 0.5 Mb showed low imputation accuracy enriched with immune function, such as 96.1% characterized genes in regions of chromosome 10, with more attention being paid on

Zhuangbiao Zhang, Ao Wang and Honghong Hu contributed equally to this work.

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downstream immune-related analysis. Our study provides the workflow of imputing lcWGS data and establishes the first high-quality cattle reference panel with free access, which provides a resource to conduct subsequent large-scale genome-wide association studies and genomic selection.

Keywords

cattle, imputation, lcWGS, phasing, reference panel

INTRODUCTION

With the development of sequencing technology, whole-genome sequencing (WGS) has become the optimal choice for population-level study of genetic variation. It plays a key role in the discovery of new variants and fine mapping the key traits, however, large-scale high-coverage resequencing requires high cost. Therefore, low-coverage whole genome sequencing (lcWGS) emerged as a cost-effective alternative choice in sequencing technology. Genotype imputation could predict the missing genotypes or the ungenotyped markers and is a common method for obtaining genotypes with higher density using linkage disequilibrium between the predicted and known markers from the reference panel [1]. The microarray data are more prone to ascertainment bias compared with the whole genome resequencing data [2] and cannot find and evaluate new variants, which affects the accuracy of genotype imputation to a further extent. In addition, the imputation accuracy, especially in livestock, is relatively low such as correlation lower than 85% [1, 3] due to factors such as chip density and reference panel.

The emergence of lcWGS could solve the above-mentioned problems. The use of lcWGS provides higher power in genome-wide association studies (GWAS) and more accuracy in polygenic risk scores compared with the microarray data [4]. The lcWGS obtains SNPs with highest density up to base-pair in large populations with lower cost while preserving individual-specific genetic information. The basic strategy is to maximize the genetic information of whole genome sequences and sacrifice the depth of coverage but obtain a greater breadth of coverage and potentially larger sample size [5]. Therefore, several imputation software designed for lcWGS have emerged, such as Gibbs sampling procedure-based GLIMPSE [6] and QUILT [7] and Hidden Markov model-based STITCH [8].

Both microarray and lcWGS require a reference panel for genotype imputation. In general, the size of the reference panel is positively correlated with the accuracy of genotype imputation. In humans, reference panels are large and diverse, such as the 1000 Genomes Project [9], the UK10K Project [10], the Genome

of the Netherlands [11], 64976 human haplotypes from European ancestry [12], and ChinaMAP [13], so that the imputation accuracy based on them could be high. Livestock panels have been reported in pigs [14, 15] and cattle [16] but were not publicly available. The establishment of both an efficient imputation pipeline and a public reference panel on cattle is crucial for research and cattle breeding improvement.

It has been shown that the imputation of lcWGS data using QUILT has the best performance (imputation accuracy >98%) after filtering in dairy cattle in the presence of 1059 Holstein cattle as a reference panel obtained from the 1000 Bull Genomes Project [17]. However, QUILT is relatively time-consuming and may lead to a higher cost in large-scale lcWGS genotype imputation.

Here, we optimized the pipeline for large-scale lcWGS genotype imputation by assessing different software. Subsequently, a high-quality cattle reference panel with free access was established by combining published and newly generated resequencing data in cattle. The imputation accuracy using the new reference panel was evaluated. The factors affecting the imputation accuracy were also explored. The currently optimal lcWGS pipelines and a large-scale reference panel established in this study will provide an important resource for GWAS and genomic selection (GS) in cattle and other species.

MATERIALS AND METHODS

Sample collection

The overall design of this study is shown in Figure 1. In this study, 3038 cattle were sequenced and collected. We generated 273 resequencing data of cattle (62 cows were sampled from Ningxia Agriculture Reclamation Helanshan Dairy Co. Ltd and 211 bulls were sampled from He Nan Ding Yuan Cattle Breeding Company Ltd.), including 131 Simmental bulls, 97 Holstein (35 bulls and 62 cows), 20 Charolais bulls, 11 Angus bulls, 9 Limousin bulls, 4 Piedmontese bulls, and 1 Jersey bull. In addition, we also collected data from 2765 cattle worldwide of more than 125 breeds, including 1555 bulls from public databases (Table S1).

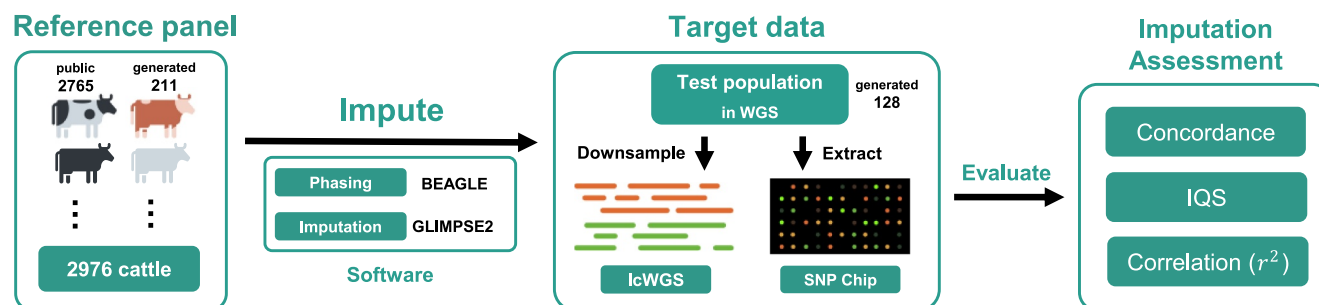


FIGURE 1 Study design. Firstly, we performed imputation to find the best phasing and imputation software and to explore the effect of diversity and size of reference panel on imputation accuracy. Then, we built a new cattle reference panel (including 2765 public samples and 211 newly generated samples) and tested it on different target data (62 Holstein and 66 Simmental cattle) to assess the performance of the new panel. Finally, we evaluated imputation accuracy through the whole genome to get a full sight of the new panel's performance. Furthermore, based on the whole genome evaluation, we used a regression model to investigate the effect of factors contributing to low imputability.

Resequencing and variant calling

The semen samples were used from the bulls for whole genome sequencing, while the blood samples were used from the cows for extractions of the DNA using the DNeasy Blood & Tissue kit (QiaGen). We used 0.5 µg genomic DNA to build a paired-end library according to the standard library preparation protocols, and sequencing was performed in DNBseq platform.

Low-quality reads were first filtered using fastp [18] with default parameters. Then, clean reads were mapped to the bovine reference genome ARS-UCD1.2 using BWA-MEM [19] v0.7.17. We used SAMtools [20] to convert and sort bam files. Then, PICARD Mark duplicates were used to exclude duplicate reads (<http://broadinstitute.github.io/picard/>). Finally, Genome Analysis Toolkit (GATK) HaplotypeCaller [21] was implemented to call SNPs. After variants calling, the GATK's variant quality score recalibration, as described in 1000 Bulls Genome Project [16] with tranche 99.9, was used to filter out low-quality SNPs, and we also excluded SNPs with missing rate >20% and minor allele count (MAC) <3 to get final SNPs databases. The average sequencing depths for 2765 cattle, 211 bulls, and 62 cows were 13.5X, 12.6X, and 16X, respectively. It should be noted that 2765 cattle and 211 bulls will be used to build reference panels, while 62 cows and 66 bulls will be the target panels. In most imputation software, they only consider biallelic SNPs; therefore, 61.8 M biallelic SNPs (Table S4) in our database (2976 cattle) were kept.

Downsample, phasing, and imputation

First, the sample coverage for each chromosome was calculated by mosdepth [22]. Then the target data for imputation was generated. For IcWGS data, SAMtools was used to downsample bam file (the average coverage is 16X) of each chromosome to 1X, 0.5X, and 0.1X. For Bovine BeadChip data, we extracted the

same SNP set as 50K and 777K chip from high-coverage sequencing data in each chromosome. Corresponding to this, we could define the genotypes from original high-coverage WGS data only at sites where depth was at least six as the true genotype and the imputation result as the imputed genotypes.

We chose Beagle 5.4 [23] and SHAPEIT5 [24] to test the speed of the phasing of the reference panel. To assess the performance of the phasing, we set $N_e = 1000$ with other parameters in default for both software. Then, we chose QUILT v1.0.4 [7] and GLIMPSE2 [25] as IcWGS imputation software and both accept bam files as an input. In addition, we also use Beagle 5.4 [26] to impute chip data based on a previous study [27] on cattle. Final imputation accuracy (see below) was shown in different minor allele frequency (MAF) bins, and running time was recorded as minutes per individual. It should be noted that all running time assessments were conducted on a Linux Server (Intel i9-12900K CPU @5.10 GHz) with six computation threads.

Metrics for imputation evaluation

Three metrics were introduced to evaluate the accuracy of genotype imputation, including concordance rate, imputation quality score (IQS), and square correlation (r^2). The concordance rate, also known as the accuracy ratio, is the ratio of the number of samples in the imputed dosage that are correct compared to the total number of samples in the true dosage. The IQS [28] was designed to evaluate the imputation reliability of a single SNP. To calculate the IQS of each SNP, we need to create a 3×3 table at first, corresponding to the genotype class aa, ab, and bb, where each cell, n_{ij} , means the number of samples with true dosage and imputed dosage (Table 1).

Based on the table, P_0 and P_c were calculated for further evaluation (Equations 1 and 2). Then, we can calculate the IQS using Equation (3).

TABLE 1 Marginal cross-classification table of true and imputed genotypes from a single SNP.

Imputed genotypes	True genotypes			Total
	aa	ab	bb	
aa	n_{11}	n_{12}	n_{13}	$n_{1.}$
ab	n_{21}	n_{22}	n_{23}	$n_{2.}$
bb	n_{31}	n_{32}	n_{33}	$n_{3.}$
Total	$n_{.1}$	$n_{.2}$	$n_{.3}$	$n_{..}$

$$P_0 = \frac{\sum_i n_{ii}}{n_{..}} \quad (1)$$

$$P_c = \frac{\sum_i n_{i.} n_{.i}}{n_{..}^2} \quad (2)$$

$$IQS = \frac{P_0 - P_c}{1 - P_c} \quad (3)$$

We defined the square correlation (r^2) as the square of the Pearson correlation coefficient between the imputed and true genotype dosage for each SNP. It can be calculated through the following Equation (4), where x_i and y_i are the dosage values from the imputed and true genotypes for the i th sample, and N is the number of samples.

$$r^2 = \frac{[N(\sum x_i y_i) - \sum x_i \sum y_i]^2}{[N \sum x_i^2 - (\sum x_i)^2][N \sum y_i^2 - (\sum y_i)^2]} \quad (4)$$

Building high-quality reference panel

To build a high-quality reference panel, we first collected 2000 cattle from public data. We use 1X lcWGS data downsampled from 62 Holstein cows with an average coverage of 16X as the target panel. We mainly focus on the effects of diversity and size of reference panels on imputation accuracy. We designed two scenarios. Firstly, a reference panel with 500 cattle was built to explore the effects of the diversity of the reference panel on imputation accuracy. Then, eight reference panels from 500 to 2000 cattle were designed to evaluate the effects of both diversity and size on lcWGS imputation accuracy.

Based on the results mentioned above, we collected 2765 cattle from previous studies (Table S1), and we also newly generated 211 bulls to increase the reference panel size. In total, 2976 cattle were included in our multi-breed reference panel.

Assessing the performance of the reference panel

To explore the performance of the newly built reference panel, the imputation was implemented on target data of three chromosomes (chr9, chr20, and chr29) from previously generated lcWGS and SNP chip data. To fully assess the imputation accuracy on the whole genome level in different breeds, we imputed the genotypes of 62 Holstein and 66 Simmental cattle. After imputation, the concordance rate, IQS, and r^2 were calculated to assess imputation accuracy.

The factors contributing to low imputability regions

To identify the genome regions that are difficult to be imputed to some extent, the low imputability regions were defined based on the squared correlation (r^2) of each SNP. The regions were defined according to the following three rules by scanning chromosomes one by one: (1) there should be at least two SNPs in a single region, (2) the r^2 of each SNP in a region should be <0.6 , and (3) the average r^2 of all SNPs in a region should be less than or equal to 0.5. Additionally, to highlight the hard-to-impute regions that are relatively longer (>0.5 Mb), we calculated r^2 for 10 Kb windows on chromosomes and defined consecutive combinations of windows with an average $r^2 \leq 0.8$ as low imputation accuracy regions, of which there must be no more than three consecutive windows with $r^2 > 0.8$ in a single region.

To further investigate the factors contributing to low imputability (Table S2), the following data were collected: (1) We used the observed heterozygosity as the indicator of the allele composition and the p -value of the HWE test for Hardy–Weinberg disequilibrium. Both of them were obtained using PLINK v1.90 based on imputed variants data [29]. (2) Mapability values were calculated based on the reference genome using GenMap v1.3.0 [30]. (3) For the marker density, according to the position of the imputed SNPs, we calculated the distance (Kb) to the nearest marker of each SNP. (4) Segmental duplication regions were identified on the reference genome by BISER v1.2.3 [31] and then filtered using thresholds by Vollger et al. [32] (5) As the indicator of conservation, we obtained mammalian ultraconserved elements (data not shown) with identity $\geq 80\%$ across all species and 100% identity in more than 3 species from our 95-way placental mammal alignment. (6) Recombination rate data were obtained from the previous study [33]. (7) The guanine-cytosine (GC) content, commonly defined as the percentage of base G or C within 25 bp upstream and downstream of a given position on the chromosome, was calculated based on the reference genome. (8) Repeat

classification data were obtained from NCBI and generated using RepeatMasker, while every repeat class was considered a single factor in the following study. (9) The structural variant regions were obtained through the following two parts: First, based on HiFi reads data from 17 cattle samples, including 10 samples collected and 7 samples from previous study, the structural variants (SVs) were detected using three methods, including pbsv v2.6.2 (<https://github.com/PacificBiosciences/pbsv>), cuteSV v1.0.13, and Sniffles v1.0.12a. Then, we collected WGS data from 401 cattle, including 10 samples collected and 391 samples from the previous study (Table S3) and genotyped SVs in these samples using Paragragh v2.4a [34]. Filtering and quality control were also carried out to ensure the credibility of the SV collection, and a more detailed processing pipeline could be found in the previous study [35]. (10) The homopolymer regions were identified based on the reference genome using software umiVar ver.2023_03 (<https://github.com/imgag/umiVar2>).

We used the multivariate logistic regression model from scikit-learn v1.2.2 [36] to explore the effect of the factors mentioned above on low imputability. Based on the *p*-values and odds ratio (the exponential of coefficient) obtained after the model's fitting calculations, we can quantify the magnitude of these effects.

RESULTS

The performance of phasing and imputation software

Previous studies have shown that QUILT works well for lcWGS data [17] but is slower when the number of animals in reference panel and sample size from lcWGS is large. To further improve the imputation efficiency, in this study, the phasing software Beagle and SHAPEIT5 were selected for running time comparison, and the imputation software QUILT and GLIMPSE2 were tested for the imputation speed and accuracy. Firstly, 500 Holstein cows were used as a reference panel for phasing in chr9, chr20, and chr29 using Beagle and SHAPEIT5. The results showed that Beagle was 1.5–2.1 times faster than SHAPEIT5 (Figure 2A–C). Secondly, it was revealed that GLIMPSE2 was nearly 3.5 times faster than QUILT with the same number of imputed variants as the reference panel. When phasing using Beagle, the trends of QUILT and GLIMPSE2 in concordance rate, IQS, and r^2 are generally consistent and of almost the same accuracy (Figure 2D–F). When phasing using SHAPEIT5, the trends of concordance, IQS, and r^2 of QUILT and GLIMPSE2 were generally consistent and of comparable accuracy (Figure 2G–I). Overall, Beagle and SHAPEIT5 have the same imputed variants and

almost the same accuracy when using the same imputation software. But there was a large difference in the time consumption between different phasing and imputation software. Therefore, considering the accuracy and efficiency, it was concluded that the combination of Beagle for phasing and GLIMPSE2 for imputation worked best for lcWGS data.

The effects of diversity and sample size of reference panel on imputation accuracy

This study further explored the effects of diversity and size of the reference panel on imputation accuracy. First, 1000 cattle of six breeds were selected from public data, including 500 Holstein, 110 BrownSwiss, 100 Jersey, 100 OriginalBraunvieh, 100 Angus, and 90 Simmental cattle. Initially, it was explored whether diversity affects the imputation accuracy when the same reference size (500 cattle) is used. The results showed that the accuracy was lower when imputing with a reference panel that did not contain the population or breeds to be imputed. When the reference group contains a certain number of populations to be imputed, increasing diversity within the reference panel can improve the imputation accuracy to some extent (Figure 3A–C). However, the accuracy decreased slightly when within-breed reference panel was used. Then, compared with using only a certain number of Holstein for the reference panel (500 cows), the accuracy gradually improved as the diversity and size of the reference panel increased. It can be seen that the accuracy was highest when the size of the reference panel was 2000 (Figure 3D–F). As a result, when building a reference panel, the size of the reference panels should be as large as possible.

Assessing the imputation accuracy of lcWGS and chip data using new multi-breeds reference panel

Considering the lack of a large publicly accessible reference panel of cattle, a new reference panel of 2976 cattle was built, including 2765 cattle from the public database and 211 newly sequenced bulls (Table S1). Given the composition of the reference panel (the biggest number of Holstein cattle in our reference panel), this study first investigated the imputation accuracy using a newly constructed reference panel for the lcWGS in Holstein cows. As demonstrated in Figure 4, the concordance rate tended to decrease with the increase of MAF, for lcWGS of 1X, 0.5X, and 0.1X, and SNP BeadChip data of 777K and 50K (Figure 4A,D,G), while the IQS and r^2 tended to increase with the increase of MAF, for lcWGS of 1X, 0.5X, and 0.1X, and chip data of 777K and 50K (Figure 4B,C,E,F,H,I). In addition, it

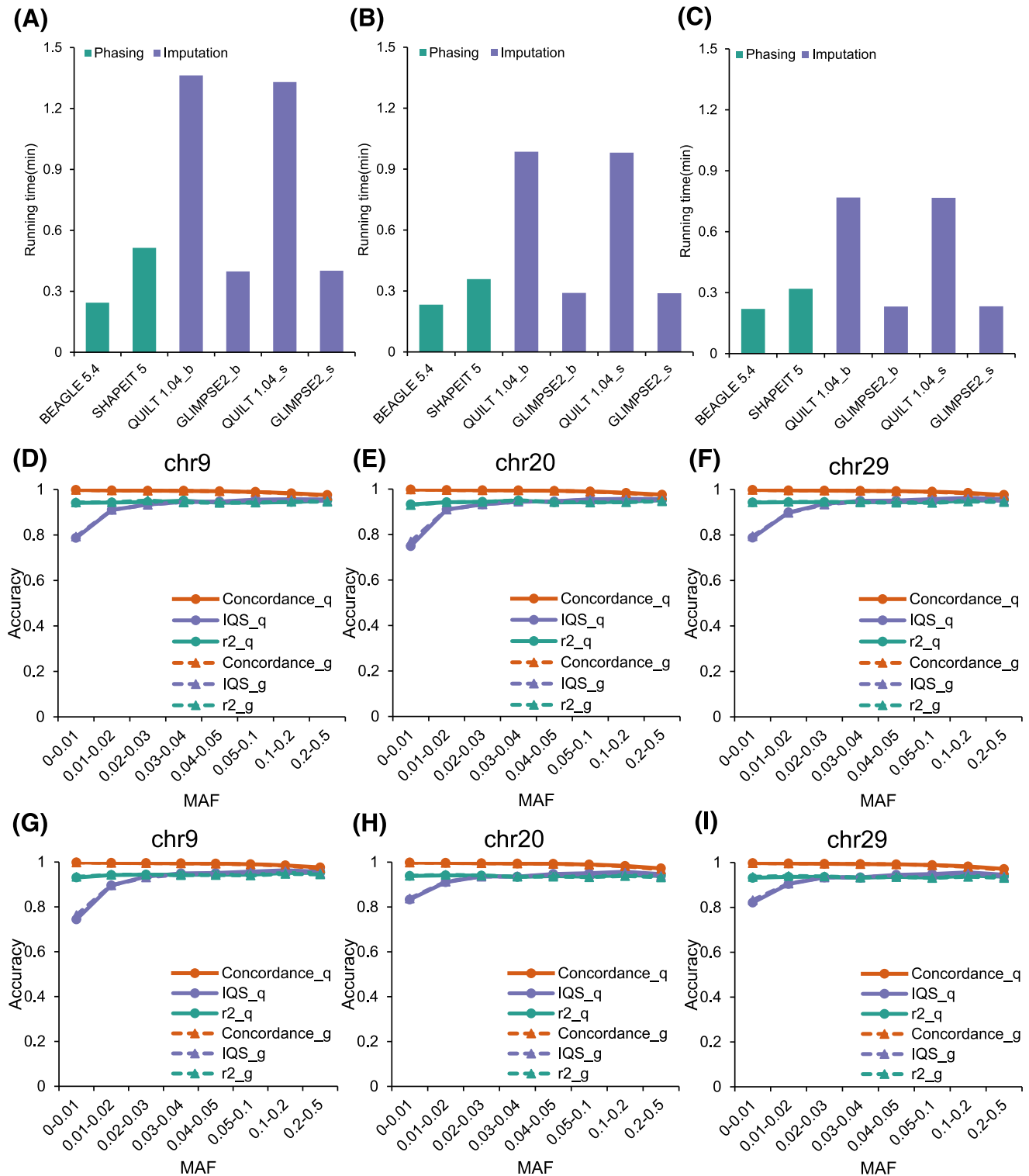


FIGURE 2 The assessment of phasing and imputation software. (A) The phasing and imputation time calculated as minutes per individual in chr9. (B) The phasing and imputation time calculated as minutes per individual in chr20. (C) The phasing and imputation time are calculated as minutes per individual in chr29. (D) The imputation accuracy of phasing using Beagle and imputation using QUILT (_q) and GLIMPSE2 (_g) in chr9. MAF: minor allele frequency calculated from reference panel, the same as below. (E) The imputation accuracy with different MAF bins of phasing using Beagle and the imputation using QUILT and GLIMPSE2 in chr20. (F) The imputation accuracy in different MAF bins of phasing using Beagle and imputation using QUILT and GLIMPSE2 in chr29. (G) The imputation accuracy in different MAF bins of phasing using SHAPEIT5 and imputation using QUILT and GLIMPSE2 in chr9. (H) The imputation accuracy in different MAF bins of phasing using SHAPEIT5 and imputation using QUILT and GLIMPSE2 in chr20. (I) The imputation accuracy in different MAF bins of phasing using SHAPEIT5 and imputation using QUILT and GLIMPSE2 in chr29.

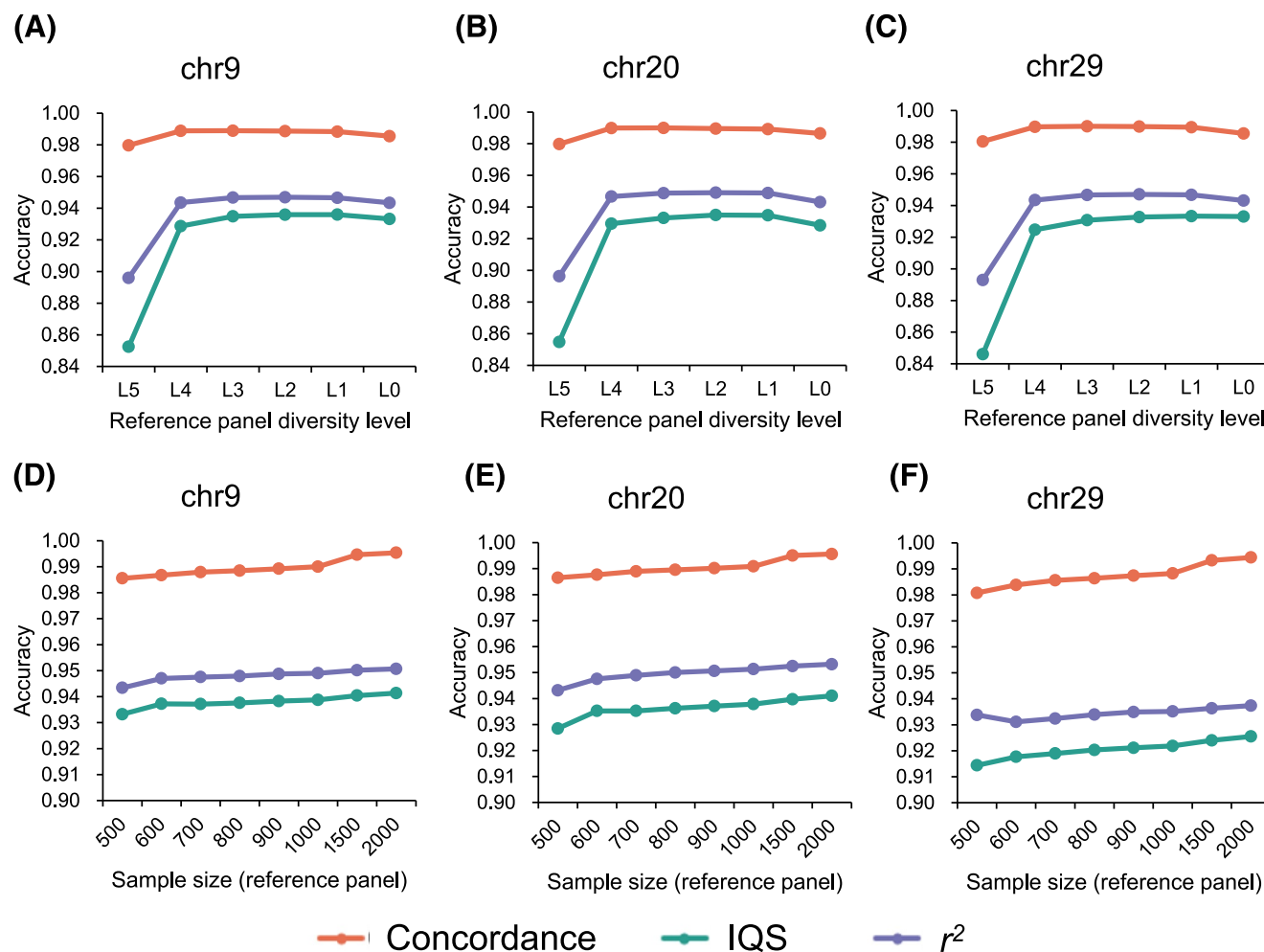


FIGURE 3 The effects of breeds diversity and size of the reference panel on the imputation accuracy. (A) The effects of the reference diversity level on the imputation accuracy on chr9, the reference panel size is 500, and the diversity level L5-0 includes breeds of five (five other breeds and no Holstein), five (100 Holsteins and four other breeds), four (200 Holsteins and three other breeds), three (300 Holsteins and two other breeds), two (400 Holsteins and one other breed) and one (only Holsteins), the same as in (B and C). (B) The effects of diversity on imputation accuracy on chr20. (C) The effects of diversity on imputation accuracy on chr29. (D) The effects of the sample size on imputation accuracy on chr9. In addition, the reference panel size is increasing from 500 (only Holstein) to 2000, the same as in (E and F). (E) The effects of the sample size on imputation accuracy on chr20. (F) The effects of the sample size on imputation accuracy on chr29. [Correction added on 12 July 2023, after first online publication: Figure 3 caption has been updated.]

should be noted that 50K SNP chip has lower accuracy than lcWGS and 777K chip data due to its low SNP density.

Our reference panel not only provides better performance for lcWGS data but also provides higher imputation accuracy for chip data, indicating that the panel was successfully established and can be used in subsequent studies.

Based on the above results, this study also performed whole-genome genotype imputation on 62 high-coverage resequencing Holstein cows (16X). The results (Figure 5A) showed that the average accuracy of imputation was 99.6% (Concordance rate: 99.2%–99.7%), 93.3% (IQS: 86.2%–94.7%), and 94.6% (r^2 : 90.6%–95.7%). In addition, the reference panel also included other cattle breeds; and to further expand the use of the reference panel, this study explored the

imputation accuracy of the new reference panel for the 1X downsample of 66 Simmental cattle (14.4X). The results (Figure 5B) showed that the average accuracy of imputation for Simmental cattle was 98.8% (Concordance rate: 95.2%–99.2%), 86.0% (IQS: 82.3%–87.9%), and 87.6% (r^2 : 84.1%–89.5%). Therefore, this reference panel has high imputation accuracy for the lcWGS data in Holstein and shows good imputation performance for other cattle breeds or populations.

The effects of different factors on low imputability regions

To further investigate the factors affecting imputability, the data of allele composition, mapability, marker density, conservation, Hardy–Weinberg disequilibrium,

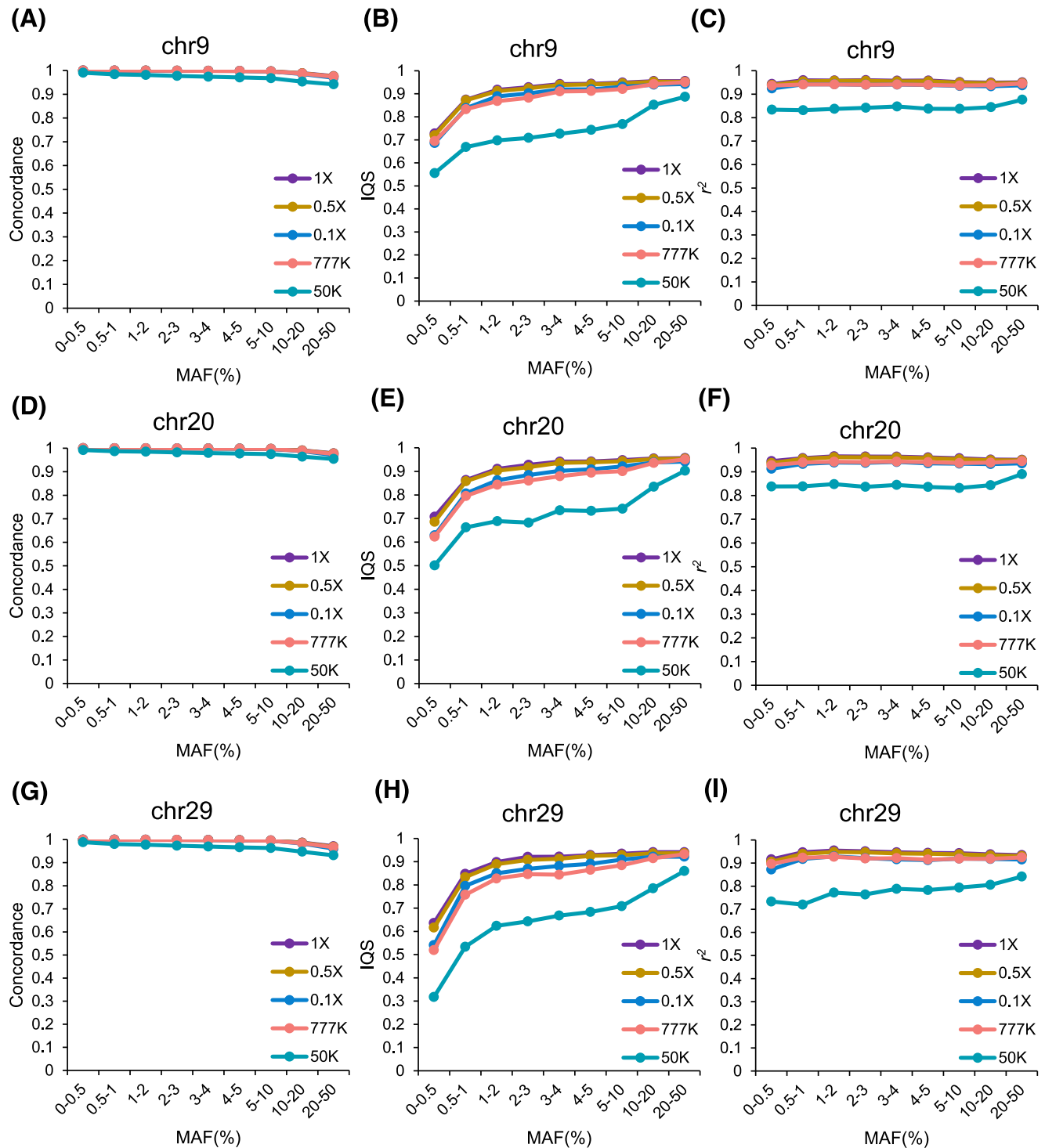


FIGURE 4 The imputation assessment of a new multi-breed reference panel with 2976 cattle. (A) The concordance rate of imputation using the new reference panel in chr9 using 1X, 0.5X, and 0.1X lcWGS data, and 777K and 50K SNP chip data, MAF: minor allele frequency. (B) The IQS of imputation using the new reference panel in chr9. (C) The r^2 of imputation using the new reference panel with 2976 cattle in chr9. (D) The concordance rate of imputation using the new reference panel in chr20. (E) The IQS of imputation using the new reference panel in chr20. (F) The r^2 of imputation using the new reference panel in chr20. (G) The concordance rate of imputation using the new reference panel in chr29. (H) The IQS of imputation using the new reference panel in chr29. (I) The r^2 of imputation using the new reference panel in chr29. IQS, imputation quality score.

recombination, GC content, repeat, segmental duplication, and structural variant were collected. We obtained 51856 low imputability regions (Figure 6A), overlapping

0.513% of the whole genome (Table S5) ranging from 556 (chr25) to 3904 (chr4) using rules defined in the method. The effects of these factors on low imputability

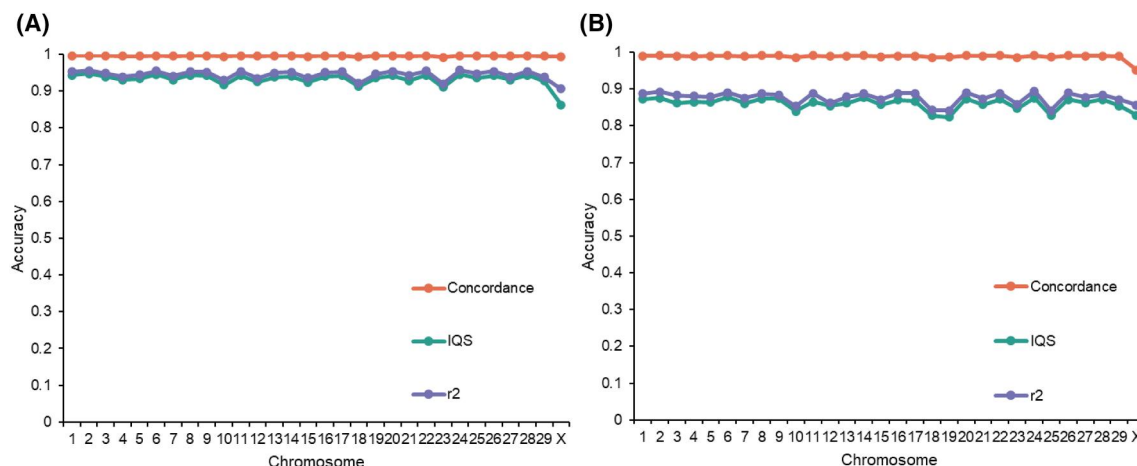


FIGURE 5 The imputation accuracy across chromosomes. (A) The imputation accuracy in Holstein cattle. (B) The imputation accuracy in Simmental cattle.

regions were investigated based on a multivariate logistic regression, showing that segmental duplication, structural variant, and GC content were the top three factors that influenced the imputability (Figure 6B,C), and SNPs with segmental duplication, structural variant, and higher GC content tend to be within low imputability regions. In addition, we also explored the whole genome and found 10 regions with a low imputation accuracy (<0.8) longer than 0.5 Mb (Table S6), such as in chr10, 12, 27, and X (Figure 6A). Previous study suggested that the regions with segmental duplication [37] were hard to impute. Our results showed that long regions with lower imputation accuracy were mainly affected by long segmental tandem repeats. In those regions, we annotated several gene families, enriched with immune function (Table S6) such as the T-cell receptor-related family in chr10, GTPase IMAP family (GIMAP) in chr4, and β -defensin family members (DEFB) in chr27. It is important to note that 96.1% characterized genes annotated in the regions of chr10 were immune related (Figure 6D).

DISCUSSION

With the meteoric development of sequencing and biostatistics technologies, the number of sequencing individuals used for genotyping continues growing to hundreds of thousands [38]. The advent of lcWGS has made it possible to genotype large-scale individuals with reduced costs and ensured accuracy. Therefore, it is crucial to construct an appropriate and efficient imputation pipeline for the current large scale data. Previous research indicated that QUILT performs the best for lcWGS data when compared with other imputation software, but its poor efficiency [17] limits imputation speed and raises the cost of imputing large-scale data. Some studies showed that pre-phasing of the target or reference panel can significantly improve the

imputation speed [39, 40]. To balance the imputation accuracy and efficiency, our study explored the imputation pipeline with phasing software using Beagle or SHAPEIT5 and imputation software using QUILT or GLIMPSE2 in cattle. Our results demonstrated that combining Beagle and GLIMPSE2 can significantly improve imputation efficiency while maintaining a high imputation accuracy. Therefore, the imputation pipeline of lcWGS data in cattle was optimized, which should also be applied to other species.

The quality of the reference panel is also important for imputation accuracy. However, there was no publicly available large-scale reference panel in cattle. In addition, the genetic diversity of the reference panel can affect imputation accuracy, such as increasing the panel diversity in humans can significantly improve microarray genotype imputation accuracy [41]. Few studies considered the breed diversity of the reference panel in livestock, whereas most studies only use a single-breed panel for genotype imputation [17, 27, 37]. Our study showed that compared to the reference panel of only 500 Holstein cattle, the reference panel, including 200 Holstein and 300 other cattle, showed the highest concordance rate; and the concordance rate increased by nearly 1% compared to other reference panels. It suggested that increasing population diversity and the reference panel size could improve lcWGS genotype imputation accuracy. Therefore, a multi-breed reference panel was established by collecting 2976 cattle data from more than 125 breeds. Compared to Teng's study [17], our imputation accuracy (IQS and r^2) in Holstein cattle are a little bit lower, while the number of samples and SNPs for chr1 in our reference panel is nearly three times larger and the rules to filter SNPs are loose. Therefore, researchers who use our reference panel for imputation can filter further according to their research purpose.

It should be noticed that some genomic regions ($r^2 < 0.8$) are difficult to be imputed. The affected

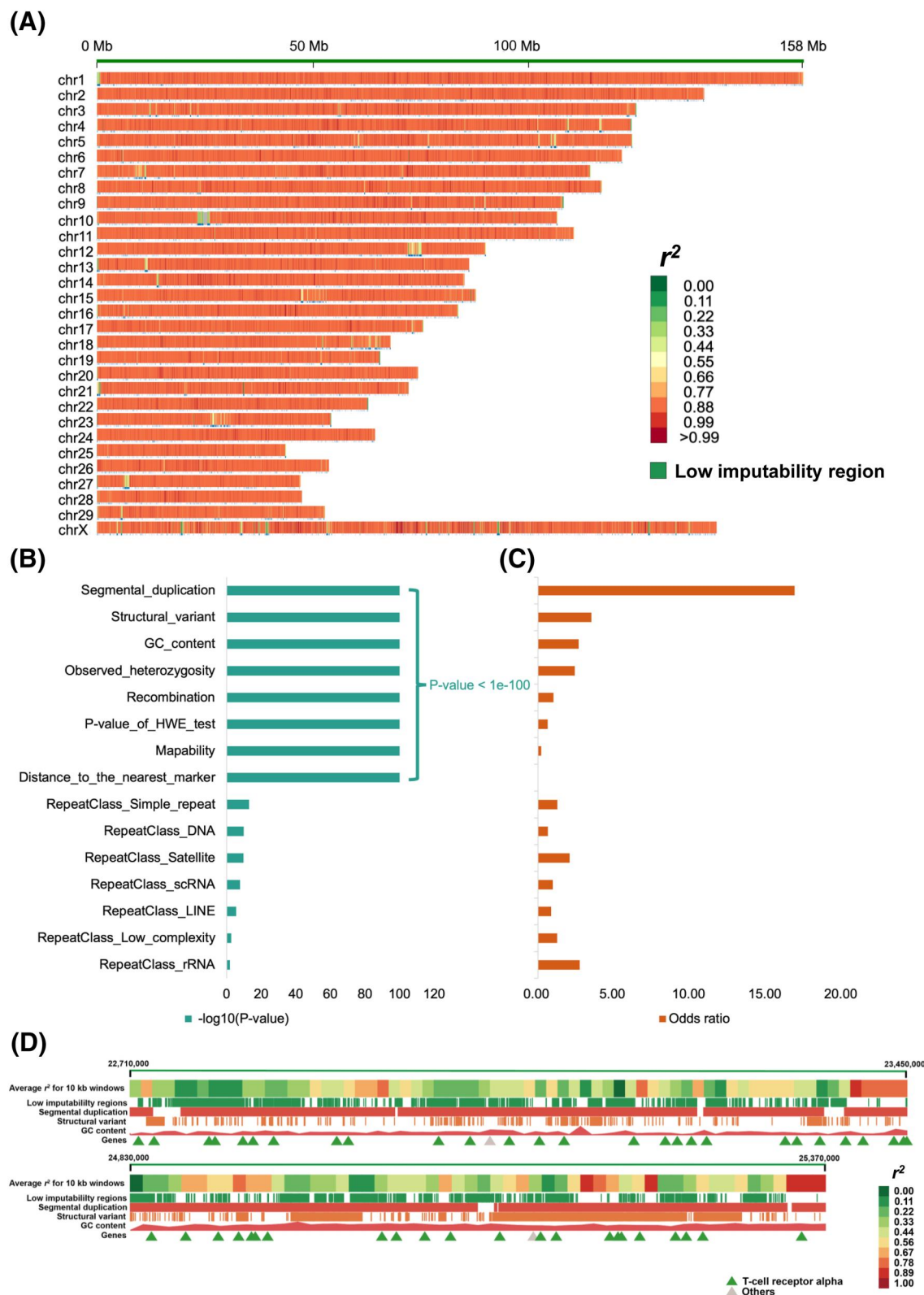


FIGURE 6 The factors attributing to low imputability regions. (A) The whole genome distribution of correlation results (first layer) in 10 Kb window and low imputability regions (second layer) plotted by RectChr (<https://github.com/hewm2008/RectChr>). (B) The p -value of multivariate logistic regression. (C) Odds ratio calculated as the exponential coefficient of multivariate logistic regression. (D) Two low imputation accuracy regions longer than 0.5 Mb in chr10 with six layers of data: (1) average r^2 in 10 Kb window, (2) low imputability regions, (3) segmental duplication regions, (4) structural variant regions, (5) guanine-cytosine content data, and (6) the distribution of genes.

factors had been studied on the imputation of microarray data in humans, which showed allele composition, Hardy–Weinberg disequilibrium, and GC content can significantly affect imputability [42]. Our study tested more than 10 candidate factors contributing to low imputability. It was found that segmental duplication, structural variant, GC content, allele composition, and recombination, etc., can significantly affect the imputation accuracy. Our study showed that recombination is not in the top three factors affecting imputability, which is consistent with the results on humans' microarray data. The top one factor in human microarray data was allele composition, while our results suggested segmental duplication is the top factor attributing to low imputability regions. The genome difference, metrics to qualify factor rank, and the method to obtain factors may cause the difference of top factors between humans and cattle. We also found several gene families in regions of low imputation accuracy longer than 0.5 Mb, such as T-cell receptor related family in chr10 (96.1% characterized immune-related genes), GIMAP in chr4, and β -defensin (DEFB) family members in chr27. Previous studies suggested that both GIMAP and DEFB are immune-related families, such as GIMAP family member could be a prognostic biomarker in breast cancer [43], and the high expression of neutrophil β -defensin in cattle is positively associated with postpartum health [44]. Interestingly, DEFB family members also demonstrated their synergistic effects on reproduction such as sperm maturation in rat epididymis [45]. Therefore, these low imputability regions may have their unique functions and masking or correcting SNPs in those low imputation regions will improve the imputation accuracy.

In conclusion, the combination of Beagle 5.4 and GLIMPSE2 was an efficient workflow for the imputation of lcWGS data in cattle. Then, the first publicly available high-quality reference panel containing 5952 cattle haplotypes was built, which performed well on both lcWGS and chip data. Finally, we found that segmental duplication, structural variant, and GC content are top three factors contributing to low imputability, which has important implications for subsequent improvement in the accuracy of genotype imputation. Overall, our work provides guidance for other species and lays an important foundation for future genomic studies such as large-scale GWAS and GS.

AUTHOR CONTRIBUTIONS

Yu Jiang: Conceptualization (lead); Funding acquisition (lead); Project administration (lead); Supervision (lead); Writing – review & editing (equal). **Zhuangbiao Zhang:** Data curation (equal); Formal analysis (lead); Methodology (lead); Resources (equal); Software (lead); Validation (equal); Visualization (lead); Writing – original draft (lead); Writing – review & editing (equal). **Ao Wang:** Data curation (equal); Formal

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CONFLICT OF INTEREST STATEMENT

The authors declare no competing interests.

DATA AVAILABILITY STATEMENT

The reference panel in each chromosome built in this study can be found at our website ([http://animal.omics.pro/code/index.php/BosVar/loadByGet?address\[\]=BosVar/Download/RefPanelData.php](http://animal.omics.pro/code/index.php/BosVar/loadByGet?address[]=BosVar/Download/RefPanelData.php)) with academic use.

ETHICS STATEMENT

All samples were taken by conforming to the Helsinki Declaration of 1975 (as revised in 2008) concerning Animal Rights, and this study was reviewed and approved by the Animal Ethical and Welfare Committee, Northwest A&F University, China.

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